

Installation Guide

TalkToJesus — A Bilingual Voice-First AI Spiritual Companion

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This document is also included as Appendix B of the final project report.

B.1 Prerequisites

Requirement	Version	Notes
Node.js	20 LTS	Production runs Node 20; development has been run on Node 24.7
npm	10+	Ships with Node 20
Flutter SDK	3.38.6 stable	Continuous integration pins this; 3.32.7 has been used locally
Xcode	15+	iOS builds only
Android Studio	Recent	Android builds only
Docker	Recent	Container builds only
gcloud CLI	Recent	Cloud Run deployment only

Accounts required: Supabase, OpenAI, ElevenLabs, Razorpay, Google Cloud with an OAuth client, and Firebase.

B.2 Getting the source

```
git clone https://github.com/manish-gitx/rn-final-manish.git
cd rn-final-manish
```

The repository is a monorepo: `TalkToJesus-backend/` is the API, `talktojesus-frontend/` is the Flutter application, and the `Dockerfile` and `cloudbuild.yaml` at the root build and deploy the backend.

B.3 Provisioning the database

In the Supabase SQL editor, run these three files **in order**:

1. `TalkToJesus-backend/supabase-setup.sql` — creates users, songs, plans, subscriptions
2. `TalkToJesus-backend/supabase-admin-setup.sql` — adds `is_admin`, `conversation_logs`, `webhook_events`, `feature_flags`, `admin_audit_log`
3. `TalkToJesus-backend/admin-bootstrap.sql` — idempotent; ends with a verification query

Running them out of order fails, because the second alters a table the first creates.

B.4 Backend configuration

Create `TalkToJesus-backend/.env`. These are the variable names the code actually reads; note that two of them differ from the names given in the repository README, and the code is authoritative.

Variable	Required	Notes
PORT	No	Defaults to 4040; the container sets 8080
NODE_ENV	No	Selects Razorpay credentials and the plan filter
LOG_LEVEL	No	Defaults to <code>info</code>
SUPABASE_URL	Yes	Module throws at import if absent
SUPABASE_KEY	Yes	Service-tier key. The README calls this <code>SUPABASE_SERVICE_ROLE_KEY</code>; the code reads <code>SUPABASE_KEY</code>
JWT_SECRET	Yes	Module throws at import if absent
GOOGLE_CLIENT_ID_WEB	Yes (≥ 1 of 3)	The README shows a

Variable	Required	Notes
		single GOOGLE_CLIENT_ID ; the code reads three separate variables
GOOGLE_CLIENT_ID_IOS	Yes (≥ 1 of 3)	
GOOGLE_CLIENT_ID_ANDROID	Yes (≥ 1 of 3)	
GOOGLE_CLIENT_ID_WEB		
RAZORPAY_KEY_ID_DEV / _SECRET_DEV / _WEBHOOK_SECRET_DEV	Yes when not production	Module throws at import if absent
RAZORPAY_KEY_ID_PROD / _SECRET_PROD / _WEBHOOK_SECRET_PROD	Yes in production	
OPENAI_API_KEY	Yes	Used for both transcription and generation
OPENAI_MODEL	No	Defaults to gpt-4o
OPENAI_MAX_TOKENS	No	Defaults to 800
OPENAI_TEMPERATURE	No	Defaults to 0.7
ELEVENLABS_API_KEY	No	If absent, replies are text- only rather than failing
ELEVENLABS_VOICE_ID	Yes for speech	
ELEVENLABS_MODEL	No	Defaults to eleven_multilingual_v 2
ADMIN_EMAIL / ADMIN_PASSWORD	Yes for the console	Console login returns 500 if unset

There is no `.env.example` in the repository; this table serves that purpose.

A note on ports. Three different ports appear across the project's documentation — 3000, 4040 and 5000. The code defaults to **4040** and the container listens on **8080**. Use those.

B.5 Running the backend

```
cd TalkToJesus-backend
```

```
npm ci
```

```
npm run dev          # nodemon on port 4040
```

Verify with `curl http://localhost:4040/`, which returns a JSON health banner. The administrative console is then at `http://localhost:4040/admin`.

Other commands:

```
npm run build        # tsc, also the CI typecheck gate
```

```
npm start            # node dist/index.js
```

```
npm test             # 66 tests, no credentials required
```

```
npm run seed:demo    # ~200 synthetic conversations across 30 days
```

```
npm run seed:demo:clear
```

`npm run seed:demo` writes clearly-marked demonstration rows: addresses end `@demo.talktojesus.local` and seeded replies carry a marker. It is demonstration data, not traffic.

B.6 Client configuration

Three files are required and are deliberately not in source control:

- `talktojesus-frontend/android/app/google-services.json`
- `talktojesus-frontend/ios/Runner/GoogleService-Info.plist`
- `talktojesus-frontend/lib/firebase_options.dart` — generated by `flutterfire configure`

For a release Android build you also need `android/key.properties` and a keystore; without them the build falls back to unsigned.

B.7 Running the client

```
cd talktojesus-frontend
```

```
flutter pub get
```

```
flutter run                                                  #
```

against the deployed backend

```
flutter run --dart-define=API_BASE_URL=http://localhost:4040 #
```

against a local backend

`API_BASE_URL` is the only compile-time define that takes effect. The others named in `EnvironmentConfig` are inert, because `main.dart` initialises the flavour with literal values instead of reading them.

B.8 Building for release

```
flutter build apk --release
flutter build appbundle --release
flutter build ios --release
```

Before submitting to Google Play, change the application identifier from the scaffold default `com.example.talktojesus` (defect D-09).

B.9 Container build and deployment

From the repository root — the `Dockerfile` expects `root` as its build context:

```
docker build -t talktojesus-backend .
docker run -p 8080:8080 --env-file TalkToJesus-backend/.env
talktojesus-backend
```

Deployment is Cloud Build to Artifact Registry to Cloud Run, service `talktojesus-backend` in `us-central1`. Note the caveat in Section 4.2.2: the live trigger currently carries its configuration inline and ignores the checked-in `cloudbuild.yaml`.

B.10 Administrative access

The console at `/admin` accepts the `ADMIN_EMAIL` and `ADMIN_PASSWORD` pair and issues a token carrying a `console-administrator` claim. That principal is synthetic — it has no row in `users`. The consequence is that there is no `is_admin` flag to clear, so **rotating the password is the only revocation, and tokens already issued remain valid until they expire.**

To grant in-application administrative access to a real user instead, set `is_admin = true` on their `users` row.

B.11 Troubleshooting

Symptom	Cause	Resolution
Backend exits at startup	A required variable is	Check <code>SUPABASE_URL</code> ,

Symptom	Cause	Resolution
	missing	SUPABASE_KEY, JWT_SECRET and the Razorpay set
503 with SCHEMA_NOT_PROVISIONED	Admin migration not run	Run <code>supabase-admin-setup.sql</code>
Every login fails	No Google client ID configured	Set at least one of the three
<code>/admin</code> returns 404 as an admin	The admin gate returns 404 by design	Confirm <code>is_admin</code> is exactly <code>true</code> , or use the console credentials
Console login returns 500	ADMIN_EMAIL / ADMIN_PASSWORD unset	Set both and restart
Replies have no audio	ElevenLabs key missing, quota exhausted, or <code>tts_enabled</code> false	Check the key and the flag
Client cannot reach a local backend	Device cannot resolve <code>localhost</code>	Use the host's LAN address in <code>API_BASE_URL</code>
Payment succeeds, app still locked	Webhook not delivered	Check the webhook secret and the console's webhook tab